

JB Fueston

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SKILLS

CAD & FEA: Autodesk Inventor and Fusion 360, NASTRAN, CATIA v5

Programming & Analysis: MATLAB, Signal Processing, Data Analysis, System Validation

Manufacturing & Hardware: Composite Layup, FDM 3D Printing, Electric Drivetrains, Suspension & Braking Systems

Tools & Systems: Jira, NASA Systems Engineering Process, Technical Communication

EDUCATION

Embry-Riddle Aeronautical University Honors College

Expected May 2027

Bachelor of Science, Aerospace Engineering

Daytona Beach, Florida

Area of Concentration: Rocket Propulsion

3.23/4.00 GPA

Relevant Coursework: Thermodynamics, Rocket Propulsion, Spacecraft Systems, Wind Tunnel Lab

PROJECT EXPERIENCE

Structures Lead - Embry-Riddle Senior Design - *August 2026 to Present*

- Executing primary structural design, volume and mass allocation, and material trade studies for a **3U Space Situational Awareness (SSA) CubeSat**, maintaining 3U envelope with **over 30% mass margin**
- Developing master CAD assemblies in **Fusion 360**, integrating vendor CAD models for COTS components to verify fit, mass, and 3U envelope compliance
- Authoring structural system verification documentation and leading design reviews to satisfy PDR and CDR requirements
- Defended preliminary component selection, CAD assembly, and cost, weight, and power budgets to Embry-Riddle faculty review board at SRR

Mechanical Team Member - Embry-Riddle Lunabotics - *August 2025 to Present*

- Executing full-system chassis and drivetrain redesigns for a **NASA Lunabotics** competition rover to minimize mass and target a **20% improvement** in excavation performance
- Maintained strict compliance with NASA systems engineering requirements across all design reviews
- Troubleshoot motor control faults to return rover to operational state
- Selected, installed, and validated upgraded gearboxes to increase drive torque by 400%
- Designed and executed drawbar pull testing to characterize tire-regolith traction

Fabrication Lead - Project Aeolus - *2022 to 2023*

- Engineered a custom wireless charging and communication uplink pad for **uncrewed aerial systems (UAS)**
- Fabricated a lightweight fiberglass composite enclosure to meet structural and mass constraints
- Authored successful technical grant proposals, securing **\$5,000** in Fort Meade research funding

Director of Fabrication, Founding Member - South River Solar Hawks - *2021 to 2023*

- Founded and directed a **20+ member engineering team** in designing, fabricating, and integrating two custom solar-powered competition vehicles
- Oversaw composite layup of full-scale **carbon-fiber aerodynamic body shells**, vehicle chassis, braking, and steering geometry
- Secured over **\$100,000** in AACPS grants and deployed **\$11,000** in tooling capital to execute vehicle fabrication and track testing for the **Texas Instruments Solar Car Challenge**

AWARDS

ERAU Presidential Scholarship

Fall 2023 to Present

Chick-fil-A Remarkable Futures Scholarship

Spring 2024